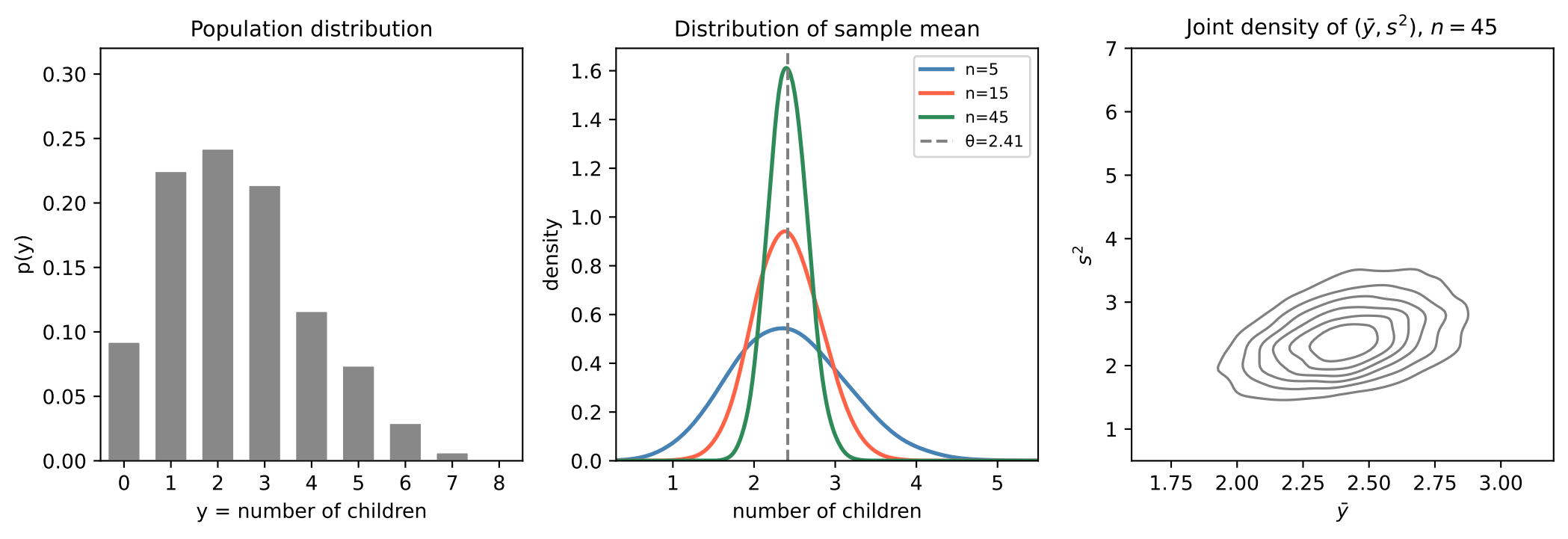
Population distribution
p(y)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
4
5
6
7
8
y = number of children
Distribution of sample mean
density
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
number of children
n=5
n=15
n=45
θ=2.41
Joint density of $(\bar{y}, s^2)$, $n = 45$
s^2
7
6
5
4
3
2
1
1.75
2.00
2.25
2.50
2.75
3.00
$\bar{y}$